Unit: Unit 2: Exponents Radicals
Topic: Exponents

Name: ____________
Date: ________

Advanced (Habanero)

Q1. Simplify $2^3 \cdot 2^4$
(A) 2^{3+4}
(B) $2^{3\cdot 4}$
(C) 2^{3-4}
(D) $2^{3/4}$
(E) $2^{4/3}$

Q2. What is $4^2 \div 4^3$?
(A) 4^{2-3}
(B) 4^{2+3}
(C) $4^{2\cdot 3}$
(D) 4^{3-2}
(E) $4^{6/2}$

Q3. Simplify $(3^2)^4$
(A) 3^{2+4}
(B) $3^{2\cdot 4}$
(C) 3^{2-4}
(D) $3^{4/2}$
(E) $3^{2/4}$

Q4. What is $x^2 \cdot x^5$?
(A) x^{2+5}
(B) x^{2-5}
(C) $x^{2\cdot 5}$
(D) $x^{5/2}$
(E) $x^{2/5}$

Q5. Simplify $(2^3)^2$
(A) 2^{3+2}
(B) 2^{3-2}
(C) $2^{3\cdot 2}$
(D) $2^{2/3}$
(E) $2^{3/2}$

Q6. What is $9^2 \div 9^3$?
(A) 9^{2-3}
(B) 9^{2+3}
(C) $9^{2\cdot 3}$
(D) 9^{3-2}
(E) $9^{6/2}$

Q7. Simplify $(8^2)^3$
(A) 8^{2+3}
(B) 8^{2-3}
(C) $8^{2\cdot 3}$
(D) $8^{3/2}$
(E) $8^{2/3}$

Q8. What is $x^4 \cdot x^2$?
(A) x^{4+2}
(B) x^{4-2}
(C) $x^{4\cdot 2}$
(D) $x^{2/4}$
(E) $x^{4/2}$

Open Ended Questions (FRQ)

Q9. Prove the rule $a^m \cdot a^n = a^{m+n}$ using examples

Q10. Simplify the expression $\frac{(2^2)^3}{2^4}$ and provide the reasoning behind your steps

Chef's Tips

- Tip 1: Ensure students apply the rules correctly and provide proper reasoning
- Tip 2: Encourage students to use examples to prove the exponent rules
- Tip 3: Remind students to simplify the expressions correctly